

MidSemester Exam Solution

EEE/ECF F311 Communication System

06-10-2025

Q. 1

$$(a) \quad h(t) = \frac{1}{2} \delta(t-1) \quad (3)$$

$$\phi = -200\pi \quad (2)$$

$$(b) \quad x_1(t) = m(t)p(t) = \frac{4}{\pi} [m(t) \cos 2\pi f_c t - \dots]$$

$$x_2(t) = \frac{4}{\pi} m(t) \cos 2\pi f_c t \quad (2)$$

$$x_3(t) = x_2(t)$$

$$x_4(t) = \frac{4}{\pi} m(t) \cos^2 2\pi f_c t$$

$$= \frac{4}{2\pi} \left[m(t) + \cancel{m(t)} \cos 4\pi f_c t \right]$$

$$= \frac{2}{\pi} [m(t) + m(t) \cos 4\pi f_c t]$$

$$\boxed{y(t) = \frac{2}{\pi} m(t) \cos 4\pi f_c t} \quad (3)$$

Q. 1

(c)

$$P_c = \frac{4^2}{2} = 8 \quad (1)$$

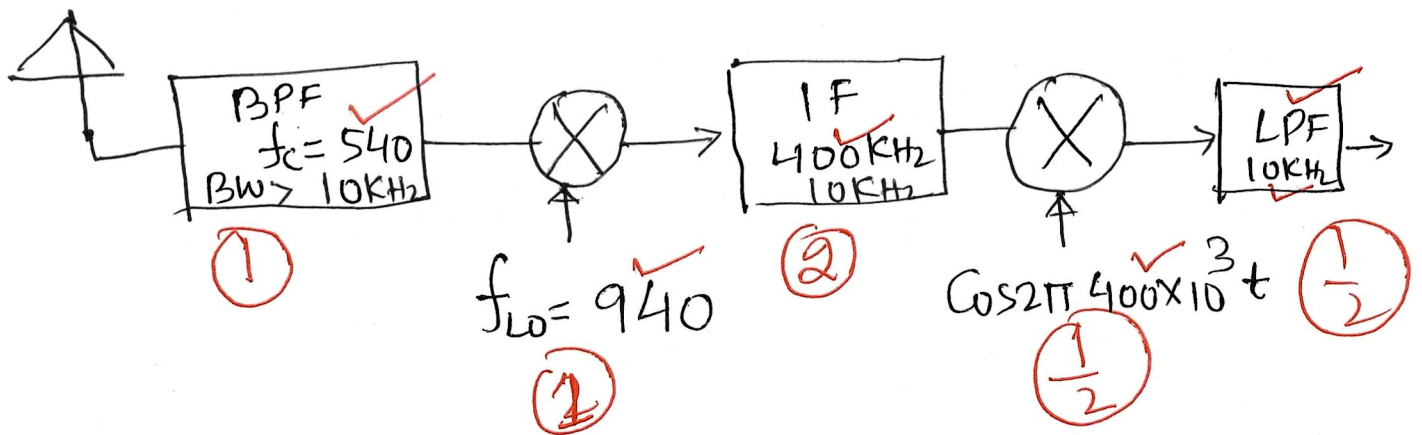
$$P_{LSSB} = \frac{P_m}{2} + \frac{P_m}{2} = \frac{8}{2} + \frac{8}{2} = 8$$

$$\eta = \frac{8}{8+8} = \frac{1}{2} = 50\% \quad (2) \quad (2)$$

(d) $f_c = 540 \text{ KHz}$

$$f_c + 2f_{IF} = 1340 \text{ KHz}$$

$$f_{IF} = 400 \text{ KHz}$$



Q. 2

(a) $m(t) = 2 \cos(200\pi t)$, $c(t) = \cos 2\pi f_c t$

$$x(t) = [A + m(t)] \cos 2\pi f_c t$$

for ED; $A = 2$

$$\therefore x(t) = (2 + 2 \cos 200\pi t) \cos 2\pi f_c t$$

$$P_{AM} = \frac{2^2}{2} + \left(\frac{2^2}{2}\right) \frac{1}{2} = 2 + 1 = 3 \text{ Watt}$$

$$P_{AM} = 3 \text{ Watt} \rightarrow (3+1) = (4)$$

(b) $\Delta f = k_f m_p = 100 \times 2 = 200 \text{ Hz}$

$$BW \equiv 2(200 + 100) = 600 \text{ Hz}$$

$$BW = 600 \text{ Hz} \rightarrow (5+1) = (6)$$

(c) $\beta = \frac{\Delta f}{f_m} = \frac{200}{100} = 2$

$$x(t) = \sum_{n=-\infty}^{\infty} J_n(\beta) \cos 2\pi(f_c + n f_m)t \quad (2)$$

$$f_c - n f_m \geq 0 \Rightarrow f_c - (\beta + 1) f_m \geq 0$$

$$\therefore f_c = 3 f_m = 300 \text{ Hz} \quad (4)$$

Q.3

(a) $L=4, m_p=4$

$$m(t) = 2000t \quad 0 \leq t \leq 2 \text{ ms}$$

$$P_m = \frac{1}{2 \times 10^{-3}} \int_0^{2 \times 10^{-3}} (2000)^2 t^2 dt$$

$$= \frac{16}{3}$$

$$SQNR = 3L^2 \frac{P_m}{m_p^2} = 3 \times 16 \times \frac{16}{3 \times 16} = 16$$

$$\therefore SQNR_{dB} = 12.04 \text{ dB}$$

(2)

Bit Stream

01 00 01 11 10

01 00 10 11 10

(2)

(b)

PCM

$$t_s = 1 \text{ ms} \Rightarrow f_s = 1000 \text{ Samples/second.}$$

$$N = 2 \text{ bits/sample}$$

$$\text{Data rate for PCM } R_b = 2000 \text{ bps}$$

$$\text{Data rate for DM } R_b = 4000 \text{ bps}$$

$$f_s = 4000 \text{ Samples/second (2)}$$

$$\max \frac{dm(t)}{dt} = 4000$$

$$\Delta = \frac{4000}{4000} = 1 \text{ (3)}$$

$$\text{or } \max \frac{dm(t)}{dt} = 2000 \text{ (1)}$$

$$\text{or } \Delta = \frac{2000}{4000} = \frac{1}{2} \text{ (2)}$$

$$\frac{\Delta}{t_s} \geq \max \frac{dm(t)}{dt}$$

Q.3

(C) Data rate for DM $R_b = 4000 \text{ bps}$
 $M = 64$, $b = 6 \text{ bits/Symbol}$

$$R_s = \frac{R_b}{b} = \frac{4000}{6} \text{ Symbols/second}$$

$$= 666.67 \text{ Symbols/second}$$

$$BW = \frac{R_s}{2} = 333.33 \text{ Hz}$$

(3)

(4)

Q.4 (a)

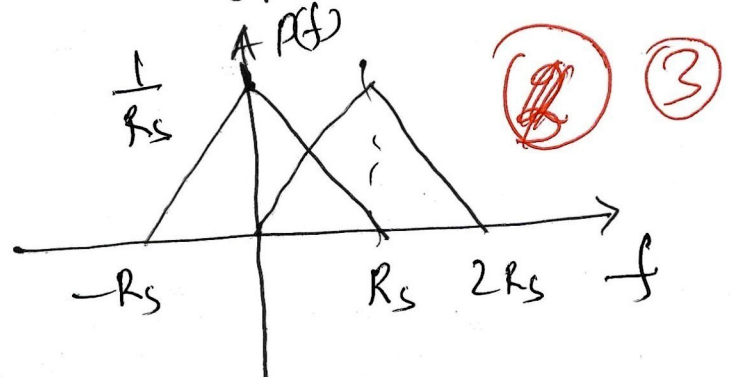
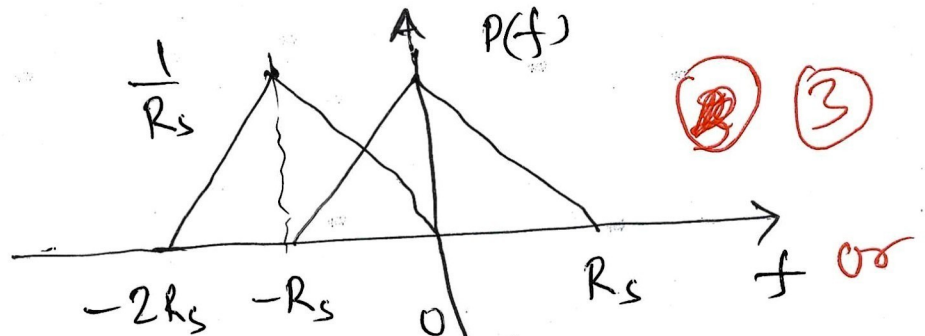
$$P(f) = \text{Sinc}(\pi R_s f) \text{Sinc}(\pi R_s f)$$

Plot

(A)

Or

(B)



$$BW \text{ of Pulse} = R_s$$

$$\text{Symbol rate} = R_s$$

$$BW = \frac{(1+\alpha)R_s}{2} \Rightarrow \alpha = 1$$

$$\therefore \text{Effective roll-off factor} = 1$$

(5)

Q.4

(b)

$$R_b = 6400$$

(2)

$$R_s = \frac{6400}{4} = 1600 \text{ Symbols/Sec.}$$

$$BW = (1 + \alpha_1) \frac{R_s}{2} + (1 + \alpha_2) \frac{R_s}{2}$$

(1)

(1)

$$2400 = (2 + \alpha_1 + \alpha_2) 800$$

$$\therefore \alpha_1 + \alpha_2 = 1$$

$$\alpha_1 = \frac{1}{2} \Rightarrow \boxed{\alpha_2 = \frac{1}{2}}$$

(6)